Geometric Sampling Strategies for Spatio-Temporal Feature Description in Video Volumes: Icosahedral versus Parallel–Meridian Representations

Harshit & Sami S. Brandt
Audio-Visual Computing, Data Science Section, IT University of Copenhagen
hano@itu.dk, sambr@itu.dk

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INTRODUCTION

Video data inherently contains both spatial and temporal information, which can be represented as a three-dimensional spatio-temporal volume with coordinates (x, y, t).

While conventional feature extraction methods generally detect and describe features independently in the individual image frames, a volumetric representation enables the direct analysis of local structures evolving through time.

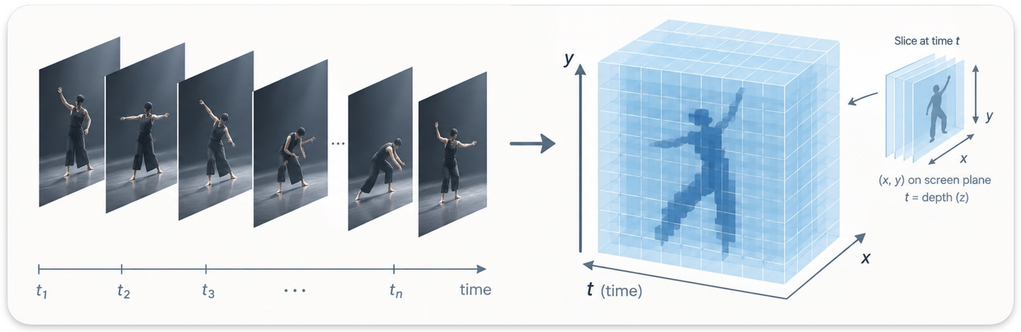


Figure 1: Video frames to voxel volume conversion

Scale-space feature detection in such video volumes can identify salient events and structures across spatial and temporal dimensions, however the presentation of the local neighbourhood surrounding a detected feature remains a critical challenge.

Stable spatio-temporal descriptors can support feature matching and tracking across video sequences and provide a foundation for multi-view correspondence, geometric reconstruction, and dynamic 3D/4D scene analysis.

METHODS

Features are detected in voxelised video-volume using a 3D extension of SIFT, and for each detected feature, a local three-dimensional neighbourhood is extracted from the video-volume and its gradient structure is analysed across multiple spatial and temporal directions.

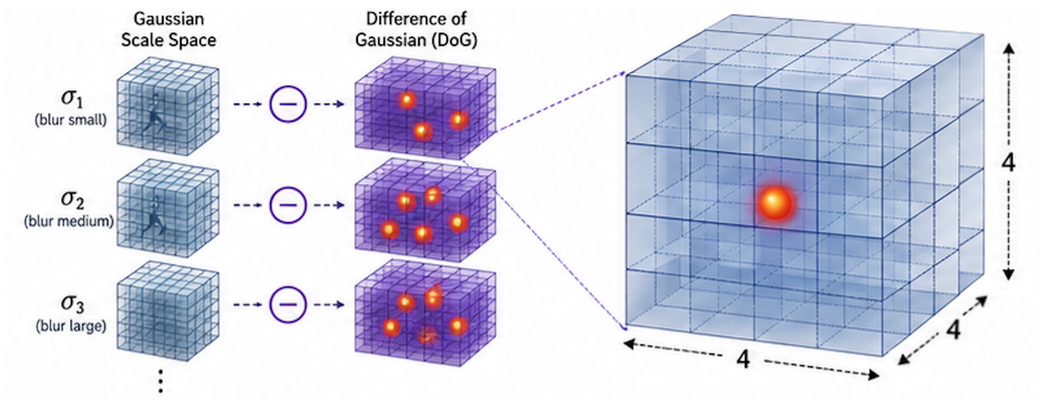


Figure 2: 3D SIFT feature detection over voxelised video volume (Scovanner et al. 2007)

The sampled directional information is subsequently aggregated into orientation histograms to construct a volumetric feature descriptor inspired by the principles of 3D scale-invariant feature representation.

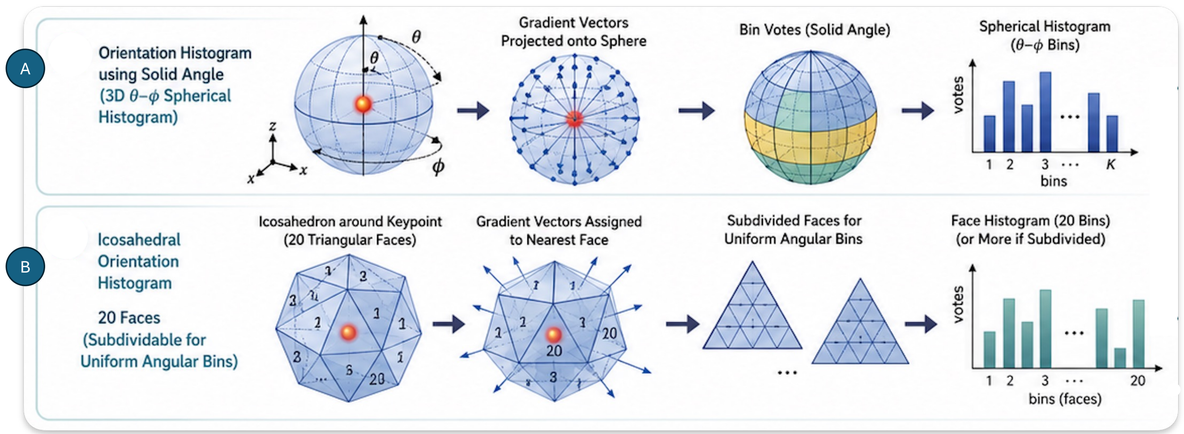


Figure 3: Descriptor Computation based on a) Parallel-Meridian , b) Icosahedral Orientation Histogram (Kläser et al., 2008)

EXPERIMENTAL RESULTS

The two sampling strategies are examined with respect to directional uniformity, angular coverage, and robustness in 4D reconstruction for spatial and temporal variations.

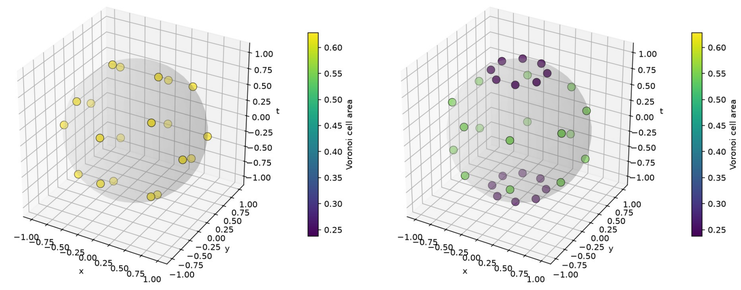


Figure 4: Spherical Voronoi cell area comparison between Icosahedron and Parallel-Meridian bin directions

Method	Points	Mean Error	Median	p90
A	2,437	9.99	10.93	12.80
B	3,068	10.51	11.14	13.37

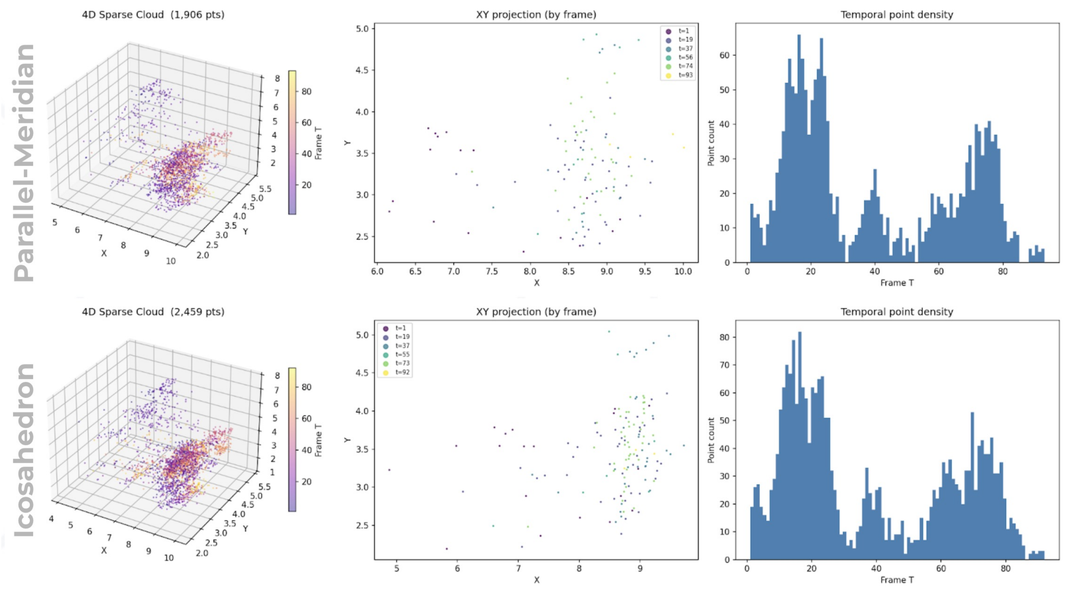


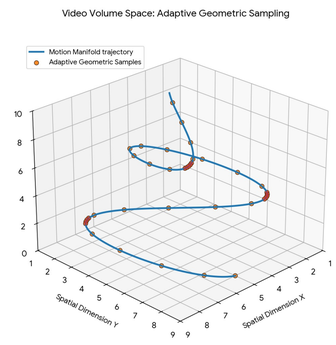
Figure 5: 4D sparse reconstruction for a camera-pair using a) Parallel-Meridian description vs b) Icosahedron description-based representation

CONCLUSIONS / FUTURE WORKS

This work investigates geometric sampling strategies for feature description in spatio-temporal video volumes by comparing two directional representations i.e., an approximately uniform icosahedral sampling scheme and a conventional spherical parallel-meridian parameterization.

The comparison addresses a fundamental question in spatio-temporal feature representation: *how does the geometry of directional sampling influence the ability of a descriptor to represent local structures embedded in space and time?*

For future work, we will explore non-Euclidean structures to sample spatio-temporal features adaptively.



By focusing computational resources on areas with high geometric complexity, we plan on achieving superior feature representation with a fraction of the data footprint.

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